

Python Fundamentals Lab

Project-Based Tasks: Build an "AI in ..." Webpage Using Python

Overview

In this lab each student (or pair) chooses one of the 15 topics below and uses Python to **generate a simple HTML webpage** about how Artificial Intelligence is used in that field. Python stores the data, processes it and writes the HTML file, so the project practises core fundamentals while producing a real, visible result.

Learning Outcomes

- Use variables, strings, lists, tuples and dictionaries to organise data.
- Apply conditionals and loops to process data and make decisions.
- Write reusable functions that return values.
- Use f-strings to build text and HTML dynamically.
- Read and write files with `open()` to save the webpage.

Tools and Time

Python 3.x with any editor (IDLE, VS Code, Thonny) and a web browser. Recommended time: 2 lab sessions (about 90 minutes each). No external libraries are required.

Common Requirements for Every Project

- The page must have a **title (h1)**, a short **introduction**, and at least one **table or card section** generated by a Python loop.
- Use at least **one function** and at least **one conditional**.
- Include a section '**How AI helps**' with 3 points written in the student's own words.
- Add simple CSS (colours, fonts, spacing) inside the Python f-string.
- Save the output as **index.html** using Python file handling.
- Add comments in the code and a README line stating the topic and student name.

Starter Template (Reference Code)

Use this skeleton for any project. Replace the data and functions with your topic. Note that CSS braces are doubled `{ { }` inside f-strings.

```
# Starter template: Python builds a webpage
title = "AI in <Your Topic>"

# 1. DATA (list / dictionary of your topic)
items = [
    {"name": "Item 1", "info": "Short description"},
    {"name": "Item 2", "info": "Short description"},
]

# 2. FUNCTION that returns HTML for one item
def make_card(item):
    return f"<div class='card'><h3>{item['name']}</h3><p>{item['info']}</p></div>"

# 3. LOOP to build the page body
cards = ""
for item in items:
    cards += make_card(item)

# 4. HTML TEMPLATE using an f-string
html = f"""<!DOCTYPE html>
<html>
<head><title>{title}</title>
<style>
    body {{ font-family: Arial; margin: 40px; background: #f4f8fb; }}
    h1 {{ color: #1F3A5F; }}
    .card {{ background: white; padding: 12px; margin: 10px 0;
        border-left: 5px solid #2A9D8F; }}
</style></head>
<body>
    <h1>{title}</h1>
    {cards}
</body></html>"""

# 5. SAVE the file, then open index.html in a browser
with open("index.html", "w") as f:
    f.write(html)
print("Webpage created: index.html")
```

Index of Projects

Project	Main Python Focus
1. AI in Chemistry	dictionaries, for loops, functions, f-strings
2. AI in Biology	strings, loops, counting, conditionals
3. AI in Medicine and Healthcare	lists of dictionaries, if/elif/else, loops
4. AI in Agriculture	lists, tuples, arithmetic, max() and min()
5. AI in Physics	functions, return values, math module, loops
6. AI in Education	nested lists/dicts, sum(), average, conditionals
7. AI in Environment and Climate	lists, loops, statistics, if conditions
8. AI in Astronomy and Space	dictionaries, sorting, lambda, loops
9. AI in Transportation	functions, arithmetic, loops, string formatting
10. AI in Finance and Banking	list of dictionaries, conditionals, counters, sum()
11. AI in Cybersecurity	strings, string methods, conditionals, loops
12. AI in Robotics	dictionaries, while loop, conditionals, functions
13. AI in Entertainment and Gaming	lists, filtering, list comprehension, sorting
14. AI in Sports	nested lists, loops, sum(), averages, max()
15. AI in Smart Cities	dictionaries of lists, loops, functions, file handling

Project 1: AI in Chemistry

Objective	Show how AI helps discover molecules, predict reactions and design new materials.
Data to use	A dictionary of 5 molecules (name, formula, molar mass, AI use). Example: Water, Aspirin, Caffeine.
Python concepts	dictionaries, for loops, functions, f-strings
Tasks	1. Create a dictionary storing 5 molecules with formula and molar mass. 2. Write a function make_row(name, info) that returns an HTML table row. 3. Loop through the dictionary to build a 'Molecule Explorer' table. 4. Add a paragraph listing 3 real AI tools used in drug or materials discovery.
Expected output	Page with heading, intro, molecule table and an AI-tools list.

Project 2: AI in Biology

Objective	Explain how AI predicts protein structures, analyses DNA and identifies species.
Data to use	A DNA string such as "ATGCGTACGTTAGC" and a list of 4 biology AI applications.
Python concepts	strings, loops, counting, conditionals
Tasks	1. Store a DNA sequence as a string. 2. Count A, T, G, C using a loop and a dictionary. 3. Calculate the GC percentage and use if/else to label it 'High' or 'Low'. 4. Display the counts in an HTML table with a short note on AI in genomics.
Expected output	Page showing the DNA sequence, base counts table and GC-content result.

Project 3: AI in Medicine and Healthcare

Objective	Present how AI supports diagnosis, medical imaging and patient monitoring.
Data to use	A list of 5 patients: name, age, heart rate (use fictional data only).
Python concepts	lists of dictionaries, if/elif/else, loops
Tasks	1. Create a list of 5 dictionaries for fictional patients. 2. Use if/elif/else to classify heart rate as Low, Normal or High. 3. Colour table rows green, orange or red using the category. 4. Add a section on 3 AI uses (X-ray analysis, chatbots, wearables).
Expected output	Colour-coded patient monitor table and an AI-uses section.

Project 4: AI in Agriculture

Objective	Show how AI helps farmers with crop disease detection, yield prediction and drones.
Data to use	A list of crops with expected yield (tonnes per hectare) for two seasons.
Python concepts	lists, tuples, arithmetic, max() and min()
Tasks	1. Store crop data as a list of tuples: (crop, season1, season2). 2. Calculate the percentage change for each crop. 3. Find the best and worst performing crops using max()/min(). 4. Write the results into a styled HTML table.
Expected output	Yield comparison page with a highlighted best crop.

Project 5: AI in Physics

Objective	Describe how AI analyses particle-collision data and simulates physical systems.
Data to use	Values for mass and velocity of 5 objects.
Python concepts	functions, return values, math module, loops
Tasks	1. Write functions kinetic_energy(m, v) and momentum(m, v). 2. Loop over 5 objects and compute both quantities. 3. Round results to 2 decimals and show them in an HTML table. 4. Add a short explanation of how AI is used at CERN or in weather physics.
Expected output	Calculator-style results table with formulas listed beneath it.

Project 6: AI in Education

Objective	Show how AI creates personalised learning, auto-grading and tutoring tools.
Data to use	Marks of 6 students in 3 subjects.
Python concepts	nested lists/dicts, sum(), average, conditionals
Tasks	1. Store marks in a dictionary: name -> list of 3 marks. 2. Calculate total, average and a grade (A, B, C) for each student. 3. Display a report card table for all students. 4. Add an 'AI Study Tips' section with 3 bullet points.
Expected output	Class result page with grades and a topper highlight.

Project 7: AI in Environment and Climate

Objective	Explain how AI forecasts weather, tracks pollution and monitors forests.
Data to use	Monthly temperature or AQI values for 6 months.
Python concepts	lists, loops, statistics, if conditions
Tasks	1. Store 6 monthly values in a list. 2. Compute average, highest and lowest values. 3. Flag months above a threshold as 'Alert' using if. 4. Draw simple bars using HTML div widths generated by Python.
Expected output	Environment dashboard with text bars and an alert list.

Project 8: AI in Astronomy and Space

Objective	Show how AI finds exoplanets, classifies galaxies and guides space missions.
Data to use	A list of 6 planets with distance from the Sun and number of moons.
Python concepts	dictionaries, sorting, lambda, loops
Tasks	1. Create a list of dictionaries for 6 planets. 2. Sort planets by distance using sorted() with a key. 3. Generate a 'Planet Cards' layout using a function and a loop. 4. Add a fact box on AI in telescopes (e.g., exoplanet detection).
Expected output	Space-themed page with sorted planet cards.

Project 9: AI in Transportation

Objective	Present self-driving cars, traffic prediction and route optimisation.
Data to use	A list of 5 routes with distance (km) and average speed (km/h).
Python concepts	functions, arithmetic, loops, string formatting
Tasks	1. Write travel_time(distance, speed) returning minutes. 2. Loop over the routes and compute each time. 3. Mark the fastest route using min() with a key. 4. Show a route table and a short note on autonomous vehicles.
Expected output	Route planner page with the fastest route highlighted.

Project 10: AI in Finance and Banking

Objective	Explain how AI detects fraud, scores credit and supports trading.
Data to use	A list of 8 transactions: id, amount, city.
Python concepts	list of dictionaries, conditionals, counters, sum()
Tasks	1. Create 8 fictional transactions. 2. Flag any transaction above a limit as 'Suspicious'. 3. Count suspicious transactions and total the amounts. 4. Show a transactions table with flagged rows highlighted.
Expected output	Fraud-check page with summary counts at the top.

Project 11: AI in Cybersecurity

Objective	Describe how AI spots malware, phishing and unusual network activity.
Data to use	A list of sample passwords or login-attempt counts (fictional only).
Python concepts	strings, string methods, conditionals, loops
Tasks	1. Write check_strength(pw) that tests length, digits and capitals. 2. Return 'Weak', 'Medium' or 'Strong'. 3. Run it for 6 sample passwords inside a loop. 4. Show results in a table plus 3 tips on staying safe from phishing.
Expected output	Password-strength report page with safety tips.

Project 12: AI in Robotics

Objective	Show how robots use AI for vision, movement and decision making.
Data to use	A list of 5 robots: name, type, battery %, task.
Python concepts	dictionaries, while loop, conditionals, functions
Tasks	1. Create a list of robot dictionaries. 2. Use a while loop to simulate battery drain of 10% per step until 20%. 3. Print the status at each step, then add it to the page as a log list. 4. Add a table of robot types and their AI abilities.
Expected output	Robot status page with a battery-simulation log.

Project 13: AI in Entertainment and Gaming

Objective	Explain AI in recommendations, game characters and content creation.
Data to use	A list of 6 movies or games with genre and rating.
Python concepts	lists, filtering, list comprehension, sorting
Tasks	1. Store 6 titles as dictionaries (title, genre, rating). 2. Use a list comprehension to filter items with rating above 4. 3. Sort the filtered list by rating. 4. Display a 'Recommended for You' section generated from the result.
Expected output	Recommendation page powered by a Python filter.

Project 14: AI in Sports

Objective	Show how AI analyses player performance, injuries and match strategy.
Data to use	Runs, goals or points of 5 players across 4 matches.
Python concepts	nested lists, loops, sum(), averages, max()
Tasks	1. Store each player's scores as a list. 2. Compute total, average and best match for each player. 3. Find the Player of the Series using max(). 4. Show a leaderboard table and a note on AI in performance analytics.
Expected output	Leaderboard page with the top player highlighted.

Project 15: AI in Smart Cities

Objective	Present AI for traffic lights, energy saving, waste management and safety.
Data to use	Electricity usage (kWh) of 5 city zones for 7 days.
Python concepts	dictionaries of lists, loops, functions, file handling
Tasks	1. Create a dictionary: zone -> list of 7 daily values. 2. Write average_usage(values) and apply it to each zone. 3. Identify the zone with the highest consumption. 4. Generate the page and save it using open() and write().
Expected output	City energy report page saved as index.html.

Assessment Rubric (20 Marks)

Criteria	Marks	What is checked
Correct use of Python fundamentals	6	Variables, data structures, loops, conditionals
Functions and code organisation	4	At least one function, clear naming, comments
Working webpage output	4	index.html opens correctly and shows dynamic content
Content quality (AI explanation)	3	Accurate, original 'How AI helps' section
Design and creativity	2	Readable layout, colours, extra features
Viva / demonstration	1	Student can explain their code

Submission Checklist

- ☐ Python file (.py) with comments
- ☐ Generated index.html
- ☐ Screenshot of the webpage in the browser
- ☐ Short reflection: 3 sentences on what you learned

Extension Ideas for Fast Finishers

- Take data from `user input()` instead of hard-coding it.
- Add a second page and link the two pages together.
- Store data in a CSV file and read it with Python.
- Use `try/except` to handle wrong input.